

capable of environment sensing and operates without any sort of human intervention. A human being is not required to take over the control of the vehicle at any time, nor even to be physically present in the vehicle. A fully automated car is self-aware and capable of making its own decisions.

Autonomous vehicles rely on sensors, actuators, artificial intelligence (AI) based algorithms, and powerful hardware to execute the operations. Autonomous vehicles create and maintain a 3D map of their surroundings based on sensors located in their diverse areas. Radar sensors monitor a vehicle's position on the road.

Video cameras detect the various objects on road like traffic lights, road signs, vehicles, and even pedestrians. Lidar sensors bounce pulses of light off the car's surroundings to measure their distances, detect road edges, and identify lane markings. Ultrasonic sensors detect the road and other vehicles in parking areas. AI algorithms process and send instructions to car controls for acceleration, braking, and steering.

Ariel view of autonomous vehicles

Autonomous vehicles are called by many names like robo-car, self-driving vehicle, autonomous car, and driverless car. With technology and sensors for radar, lidar, sonar, GPS, odometer, and inertial measurement unit (IMU), autonomous vehicle can be programmed quickly with autopilot capabilities.

Some of the common sensors used in an autonomous vehicle are explained below:

Radar. It uses a sensor that helps to find the range, angle, and velocity of vehicle using radio waves. It is the same technology that is used in spacecraft, guided missiles, and aircraft. It has various transmitters that produce different frequencies of radio waves, which are used to sense various parameters required for vehicle motion.

Lidar. It uses minute laser light emission like a bat that flies and identifies obstacles based on sound reflection. The laser light emitted by

lidar is reflected by the obstacles it faces and the reflection time is used to calculate their distance.

Sonar. Sonar is a sound navigation and ranging sensor that uses sound propagation to assist navigation and detection of objects through echo characteristics or acoustic location using very low frequency called infrasonic or very high frequency called ultrasonic. Sonar sensors are of two types: active sonars which emit sound pulses and listen to the echo reflection of the sound and the passive sonars which listen to sound made by different types of objects to identify them.

GPS. Global positioning system (GPS) is a satellite based radio navigation system to get geolocation and time for navigation and potential routes.

Odometer. Odometer uses a technique called Odometry that uses motion sensor data to estimate change in vehicle position and movement relative to a given point or starting location. It is commonly used in robots for leg and hand movements and has inspired use in autonomous vehicle motion.

IMU. Inertial measurement unit is a group of electronic sensors like accelerometer, gyroscope, and magnetometer that is used to measure and report various parameters like force of motion/movement, angular rate, and orientation.

Signage sensor. This is more of an image recognition system to read and interpret signage boards on the roads and feed as input for vehicle movement and direction. For example, signage boards like speed limit and diversion are interpreted and used for vehicle movement, acceleration, and direction.

Companies like Tesla and Waymo have been instrumental in introducing various new sensors and facilities to improve the behaviour of autonomous vehicles. One of the main characteristics of autopilot application is that the software built into the autonomous vehicle has the capability of reprogramming engine so that periodic updates of the software can be provided by vehicle supplier or customised by vehicle owner just like Android updates for smartphones.

One of the key problems in autopilot application and its connected sensors is susceptibility of the sensors and navigation system in specific weather conditions like snow or rainy season and also attack threats like jamming and spoofing. Another challenge can be termed as social challenge where the vehicle has to be updated frequently for high-definition maps, potential future regulations, road condition changes, and route changes.

For precision of vehicle movement and sensor handling, aerial image, high-definition views, and aerial view of sensors like lidar and odometry can help in getting 3D positions for accurate working of autonomous vehicle. For a congested route, aerial imagery maps and lidar using ground reflective position help to get accurate autopilot calculations. This is one of the recent advancements in autonomous vehicle technology that uses the combination of localisation framework of 3D lidar to create local ground reflectivity grid map using aerial localisation (AL) and lidar ground reflectivity (LGR) to enhance the facilities of autopilot applications.

Autopilot applications and their usage

Autopilot is the facility in a vehicle to handle, steer, accelerator, brake automatically based on various inputs from sensors with less or no manual/human inputs. This means we need to design a self-driving car with autopilot software application that should get the inputs from various sensors and actuators and decide how to drive based on these inputs coupled with artificial intelligence based training model in-built in the autopilot application.

Autopilot applications should work in real time to steer or accelerator or apply brake in the vehicle based on inputs received. So the training model and the processing efficiency should be super-fast to be able to provide real-time instructions to control the vehicle. The purpose of autopilot is to reduce manual effort in driving the vehicle and improve automated facilities to do the same. It can be a semi-automated vehicle